

Airfoil Laboratory

MAE 650:433 Aerospace Engineering Laboratory
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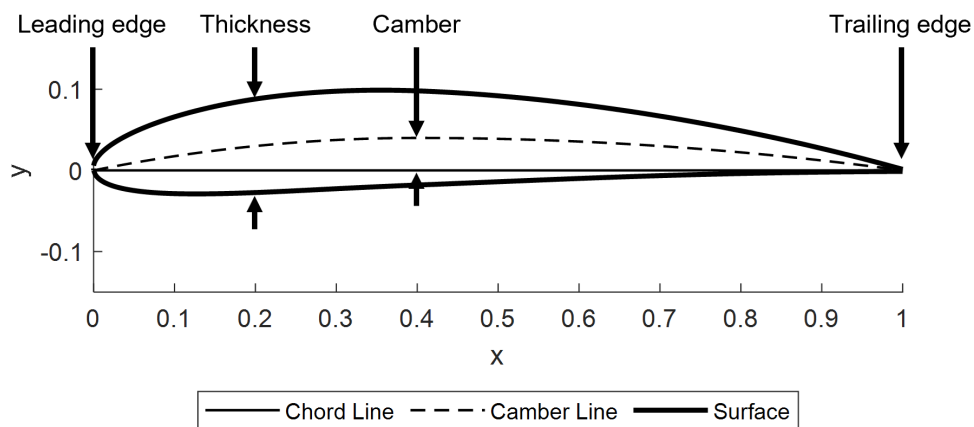
Fall, 2026

1 Introduction

With the advent of heavier than air flight by Otto Lilienthal and later the first powered flight by Orville and Wilber Wright at the beginning of the 20th century, aerodynamic researchers have been interested in the most efficient (providing the highest lift with the lowest drag) shape of wings. In 1918, Ludwig Prandtl separated the study of wings into two parts:

1. The study of the wing cross-section called an airfoil.
2. The modification of the airfoil properties along the finite wing.

The first systematic study of airfoil shapes was conducted by Horatio Phillips in 1884, and by the 1930s, the National Advisory Committee for Aeronautics, NACA, (the forerunner to NASA) systematically studied and cataloged airfoil shapes and their performance. Figure 1 defines the main characteristics of an airfoil.



- Mean camber line: locus of points halfway between the upper and lower surfaces as measured perpendicular to the mean camber line itself.
- Chord line: straight line connecting the leading and trailing edge.
- Chord: distance measured from the leading to the trailing edges along the chord line.
- Camber: maximum distance between the chord line and mean camber line.
- Thickness: distance between the upper and lower surfaces perpendicular to the chord line.

Figure 1: Nomenclature, airfoil NACA 4412.

Since the initial studies, hundreds of airfoil shapes have been evaluated and optimized depending on their application. Common applications of airfoils include lifting surfaces (wings and control surfaces), propellers, turbine and fan blades, automobile spoilers, and windmills for generating power. The performance characteristics of cataloged airfoils, including the lift, drag, pressure distribution, and moment about the aerodynamic center are also cataloged. Values for these characteristics have been measured experimentally and determined through theories and computational models with varying degrees of complexity.

2 Problem Statement

In the current laboratory, you are asked to measure the lift and drag of five different airfoils using a force balance and small subsonic wind tunnel, to compare the results with published data and theoretical analysis, and to provide arguments for the choice of particular airfoils for specific applications. Four of the five airfoils are standard NACA four-digit sections chosen so that camber, camber location, and thickness are each varied one at a time. The fifth is a deliberately non-standard section that falls outside the four-digit family, included so that the trends established by the first four can be tested against a shape they do not describe.

3 Experimental Facility

The experiments will be performed using a 6" by 6" subsonic wind tunnel equipped with a force balance. The wind tunnel is an open in-draft design (the test section is upstream of the fan), which minimizes the turbulence induced by the fan. The wind tunnel flow velocity is controlled using a sliding plate, which throttles the exhaust of the wind tunnel. A static pressure tap connected to an inclined manometer is located upstream to indicate the "uniform" flow velocity entering the test section. The wind tunnel can also be equipped with a hot-film anemometer (which has been removed for this experiment).

A force balance, see Figure 2, is used by researchers to measure the reaction forces and moments on a model and allows the orientation (angle of attack, pitch, yaw) to be changed during a test. Complex force balances in large-scale facilities can simultaneously measure all three forces and three moments on the model. The force balance in this experiment will be used to measure only the axial and normal force relative to the rotation sting, as shown in Figure 2. Using the axial and normal force, and the rotation sting angle θ , the lift and drag can be computed using the relationship shown in Figure 2. The forces are measured using the load cell, which is the hollow rectangular device at the end of the sting. The rotation sting is connected to a shaft equipped with an indexed rotation stage, allowing the angle of attack to be varied throughout the test. Since the rotation stage is stiff, only use the side-arm located on the stage outside the tunnel to rotate the model. When the thumbscrew on the rotation stage is loose, the stage can be rotated coarsely, and when it is tight, a fine rotation angle micrometer can be used for a fine adjustment. The load cell uses strain gauge circuits whose voltage output is calibrated to read the force in Newtons.

Five finite wings with different airfoil types, some shown in Figure 3, will be used for this experiment. The airfoils are attached to the load cell using a mounting sting, which is bent so that the airfoils are oriented at a slight angle of attack (ϕ_0) when the rotation sting is at zero degrees. This offset mounting sting angle is needed to provide the large airfoil angles of attack during the experiment since the force balance has a limited range of rotation. The offset angles for each airfoil are given in Table 1 so that the actual angle of attack (α) can be calculated for each rotation sting angle, as in (1).

$$\alpha = \phi_0 + \theta \quad (1)$$

Since you will be examining several different airfoils, it is useful to understand the NACA numbering conventions for the four-digit series (i.e., like NACA 0012). The first digit specifies the

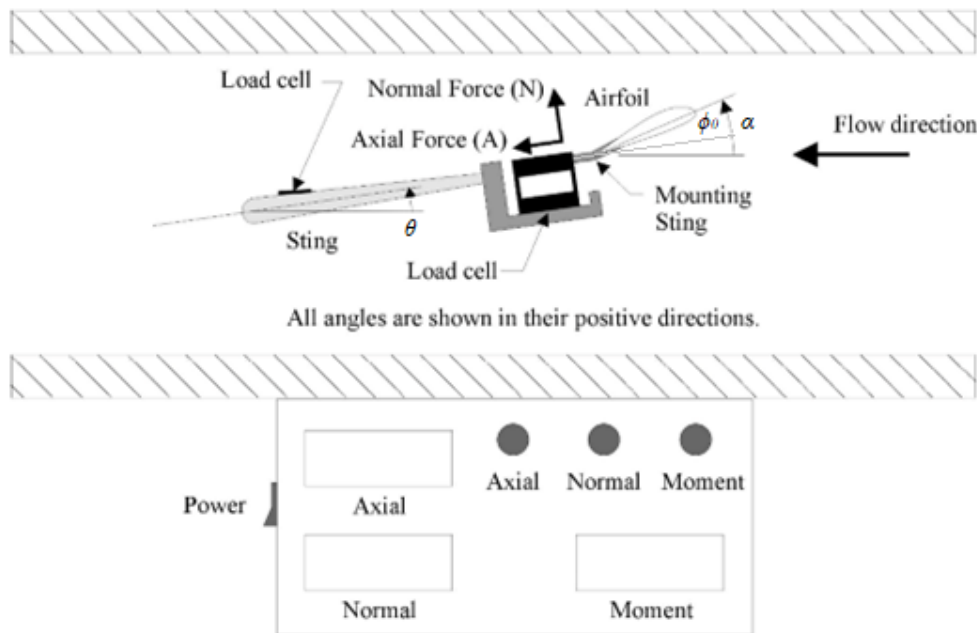


Figure 2: Schematic of wind tunnel and balance.

maximum camber (m) in the percentage of the chord (airfoil length). The second digit indicates the position of the maximum camber (p) in tenths of the chord, and the last two numbers provide the maximum thickness (t) of the airfoil in the percentage of the chord. For example, the NACA 2415 airfoil has a maximum thickness of 15% with a camber of 2% located 40% back from the airfoil leading edge (or $0.4c$). The four standard airfoils in this experiment are ordered so that each successive comparison isolates a single geometric variable, as summarized in Table 1: NACA 0012 to 2412 isolates the effect of camber, 2412 to 2212 isolates the effect of camber location, and 2412 to 2424 isolates the effect of thickness. The fifth airfoil is not a member of this family and is used as a point of contrast.

4 Experimental Procedure

Turn the power of the force balance on, which is located on the back of the readout, immediately when you arrive and let the unit warm up for approximately 5 minutes before making measurements.

Zeroing of the Sting Angle and Force Balance First, the rotation stage must be zeroed relative to the free stream direction. To accomplish this, with the empty load cell, rotate the sting and set it on zero angles. Next, the load balance must be zeroed. To do this, press the Tare button on both the axial and normal force gages.

Setting the Force Balance Mounting Sting Offset In general, when using a force balance in wind tunnels, the researcher would like to measure only the forces and moments due to the model and neglect the effects of the mounting hardware and sting that attach the model to the balance. Unfortunately, this can be difficult since the effect of the mounting sting is influenced by the presence of the model. Therefore, it is customary to use a calibration mounting sting similar to that used on the airfoils, run the tunnel with only the mounting sting attached, and record the forces for different angles of attack. Since the load cell is located at the end of the sting, the only effect in the present system is due to the shape of the load cell itself and the small mounting sting.

1. Attach the small calibration mounting sting (the threaded rod with no airfoil attached) onto the force balance and slightly tighten the setscrew. Make sure the calibration sting is oriented at a positive angle of attack when the rotation sting angle (θ) is at zero degrees.
2. With the wind tunnel off, note the wind tunnel speed manometer height.
3. Turn the wind tunnel on but close the flow control plate so there is no flow in the wind tunnel.
4. Record the axial and normal forces at each of the six rotation sting angles listed in Table 2, that is, at $\theta = -15^\circ, -10^\circ, -5^\circ, 0^\circ, +5^\circ$, and $+10^\circ$. Record these forces for “wind on” (control plate fully open so that the tunnel is at maximum velocity) and “wind off” configurations. It is faster to make all the “wind off” measurements first, and then continue to the “wind on”. For each angular position, record the sting angle, wind tunnel air velocity (differential pressure as measured by the angled manometer), and axial and normal force with the control plate opened. **Note that the load cell is calibrated to Newtons, which are directly read from the readouts.**
5. These six sting angles are the only ones used in the experiment, so the calibration sting must be run at exactly these angles. If the offset angle ϕ_0 of any airfoil differs from 10° , the corresponding sting angles must be recomputed from (1) and the calibration sting run at those angles as well.

Airfoil Axial and Normal Force Measurements All five airfoils listed in Table 1 are to be tested, using the following procedure for each one. Each airfoil is tested at the six angles of attack listed in Table 2, giving $5 \times 6 = 30$ trials in total. Each trial consists of one “wind off” and one “wind on” reading of the axial and normal force at the same sting angle.

The fifth airfoil has not yet been selected. Once it has been, enter its span, aspect ratio, and measured offset angle in Table 1 before testing, and recompute its sting angles from (1).

Table 1: Airfoil Options For Testing.

Airfoil	Span	Aspect Ratio	Offset Angle (ϕ_0)	Role
0012	5"	2	10°	Baseline, symmetric
2212	5"	2	10°	Camber at 0.2c
2412	5"	2	10°	Camber at 0.4c
2424	5"	2	10°	Thickness 24%
TBD (non-standard)	—	—	—	Contrast case

Confirm the span and chord of every model by direct measurement before testing; the values listed assume the standard 5" span, aspect-ratio-2 planform, and the wing area S and chord c taken from these measurements are used directly in (3).

Table 2: Test matrix. Rotation sting angles required for each angle of attack, $\theta = \alpha - \phi_0$.

Angle of attack, α	−5°	0°	5°	10°	15°	20°
Sting angle, θ (for $\phi_0 = 10^\circ$)	−15°	−10°	−5°	0°	+5°	+10°

1. Attach the airfoil to the force balance without tightening the mounting setscrew. Make sure the airfoil is oriented at a positive angle of attack when the rotation sting angle (θ) is at zero degrees. To ensure that the airfoil is parallel to the wind tunnel floor, increase the sting angle until the leading edge of the airfoil is almost touching the ceiling of the wind tunnel. With the loose mounting sting in the load cell, rotate the airfoil until the leading edge of both wingtips is parallel to the ceiling of the wind tunnel. Slightly tighten the setscrew to secure the airfoil to the sting.
2. Before turning on the tunnel, sweep the model through the full range of sting angles in Table 2, from $\theta = -15^\circ$ to $\theta = +10^\circ$, and confirm two things: that the rotation stage reaches both ends of the range, and that at every angle the load cell is not resting on the floor and the leading edge of the wing clears the wind tunnel floor and ceiling by at least 20mm (the boundary layer on the wind tunnel floor may affect the force measurements). If either end of the range cannot be reached, the offset angle ϕ_0 of the mounting sting must be reset, or the model inverted to reach $\alpha = -5^\circ$; record whichever was done, since it changes the sign convention of the measured forces.
3. Turn on the wind tunnel.
4. Record the axial and normal forces at each of the six sting angles in Table 2. For each position, record the sting angle, wind tunnel air velocity from the manometer, and the axial and normal forces. For each angle, record the forces with the flow control plate completely closed and a second time with the control plate completely open. Again, it is faster to first record the forces for the “wind off” configuration at the various sting angles and then continue to the “wind on” measurements for the same set of sting angles.
5. The 5° spacing of the required angles of attack is coarse near stall, and the stall angle can only be bracketed rather than resolved by these six points alone. If time permits after

all 30 required trials are complete, take additional points every 1° between $\alpha = 10^\circ$ and $\alpha = 20^\circ$ for one or two of the airfoils. Report these separately from the required 30 trials and identify them clearly as supplementary.

6. Turn off the wind tunnel. Remove the current model and proceed to the next airfoil. Repeat steps 1–5. Before you leave, make sure that you know how to convert the inclined manometer readings into windspeed.

5 Items to be Included in the Report

1. Define the following terms associated with airfoils:

- | | |
|------------------|---------------------|
| (a) Lift | (e) Lift slope |
| (b) Drag | (f) Stall |
| (c) Profile Drag | (g) Flow separation |
| (d) Induced Drag | (h) Kutta condition |

2. From the components of the force, measurements show that the lift and drag are given by (2):

$$D = A \cos \theta + N \sin \theta, \quad L = N \cos \theta - A \sin \theta \quad (2)$$

where L is the lift force, D is the drag force, A is the axial force, N is the normal force and θ is rotation sting angle.

Remember to subtract the weight of the airfoil (sting axial and normal force for the airfoil with the wind off) and the weight and lift and drag force of the force balance mounting (measured earlier in both winds on and wind off tests) to find the true (calibrated) axial and normal forces.

3. Tabulate the sting angle (θ), angle of attack (α), lift force (L) in Newtons, the drag force (D) in Newtons, the lift coefficient (C_L), and the drag coefficient (C_D). Also, calculate the Reynolds (Re) number for each airfoil. These are given by (3):

$$C_L = \frac{L}{q_\infty S}, \quad C_D = \frac{D}{q_\infty S}, \quad Re = \frac{V_\infty c}{\nu}, \quad q_\infty = \frac{1}{2} \rho_\infty V_\infty^2 \quad (3)$$

where V_∞ is the free stream velocity, S is the wing area (chord times width), c is the chord length, and ν is the kinematic viscosity. The table should contain 30 rows, one for each trial, grouped by airfoil.

Note that the mounting sting of the airfoil is at an offset angle and not aligned along the imaginary line from the airfoil leading to trailing edges. Using the sting offset angles in Table 1, the actual angle of attack of the airfoils should be calculated from the rotation sting angle (θ) and the sting offset angle (ϕ_0). Also, when calculating the lift and drag, the tunnel off and calibration mounting sting forces should be subtracted from the measurements.

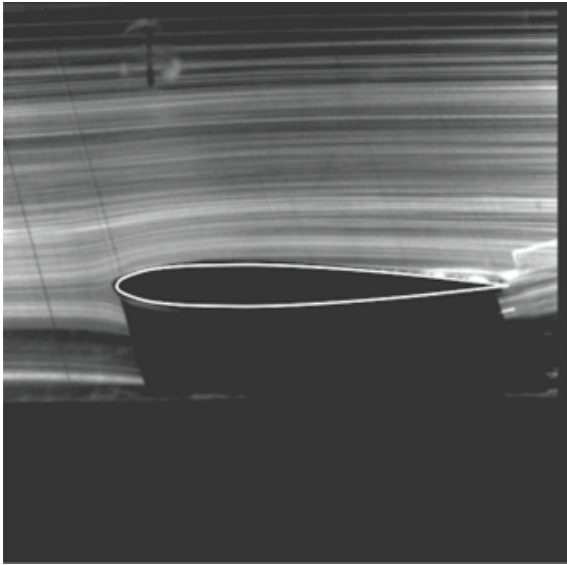
4. Plot three graphs: lift coefficient versus angle of attack, drag coefficient versus angle of attack, and the drag polar (C_L versus C_D on the x-axis). In each graph show the results for all five airfoils. Also, plot the theoretical thin airfoil theory lift coefficient for symmetric

airfoils, which is given from thin airfoil theory as $C_L = 2\pi\alpha$ (where α is in radians). Since only six angles of attack were measured, plot the measured points as discrete symbols rather than as a continuous curve, and fit the lift slope only over the points that lie in the linear region.

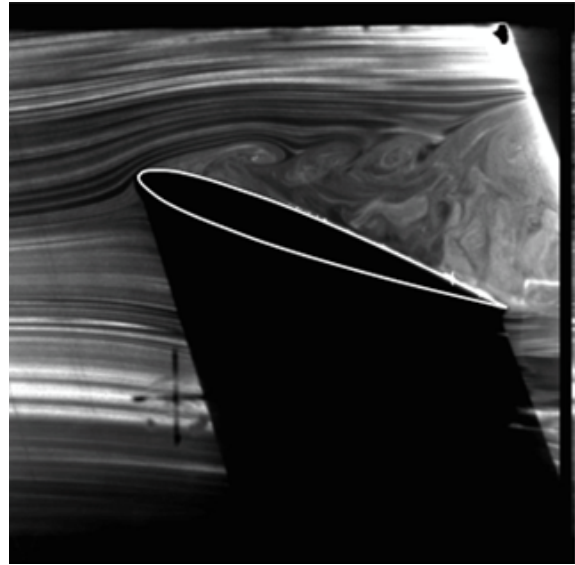
5. Discuss the differences you observe in the profiles between the airfoils and give the reasons for the differences when possible. Why do the lift and drag plots have the trends they do (i.e., what is occurring in the flow field around the airfoil causing the plot to have the shape it does? Refer to Figure 3). Address each of the three isolated comparisons in turn:
 - (a) NACA 0012 versus 2412: the effect of adding camber at fixed thickness. What happens to C_L at $\alpha = 0^\circ$, to the zero-lift angle of attack, and to the lift slope?
 - (b) NACA 2412 versus 2212: the effect of moving the maximum camber location forward from $0.4c$ to $0.2c$ at fixed camber and thickness.
 - (c) NACA 2412 versus 2424: the effect of doubling the maximum thickness at fixed camber. What do you expect to happen to the drag at low angle of attack and to the stall behavior?
6. Discuss the fifth (non-standard) airfoil separately. State why it was selected, in what way its geometry departs from the four-digit family, and which of the trends established in item 5 it follows and which it violates. If it behaves in a way the four-digit comparisons would not have predicted, explain the flow physics responsible.
7. Discuss the similarities and differences in the curves obtained from the experiment and the attached plots from [1], also in Figures 4 and 5. Note that Figure 5 is the published data for NACA 0012, which is tested directly in this experiment, while Figure 4 is the published data for NACA 4412, a cambered four-digit section that is not in the present test matrix but which serves as a reference for the effect of camber. Give possible reasons for the differences (note that the differences are not entirely due to experimental errors). What is the difference between section drag and the drag measured in this experiment? How does the finite aspect ratio affect the lift coefficient? Why does the stall behavior of your airfoils look significantly different than those given in Figures 4 and 5?
8. Comment on the angular resolution of the experiment. With measurements only every 5° , how precisely can the stall angle be located for each airfoil, and how much uncertainty does the 5° spacing introduce into the lift slope you extracted in item 4? Compare this to the other sources of uncertainty in the measurement.
9. If you were asked to select an airfoil for a model aircraft with a chord length of $0.2m$ and a velocity of $5m/s$, would it be better to use the published data given or measured data from this experiment (give an explanation for your answer)?
10. From your data, state which airfoil you would use and discuss the reasons why you selected the airfoil for the following applications.
 - (a) Aircraft at cruise conditions.
 - (b) Aircraft landing.

- (c) Racing car spoiler where the rules state that the airfoil must be at zero angles of attack.

Be sure to give sample calculations for the above analysis and discuss the accuracy of the results.



(a) Zero degrees angle of attack.



(b) 21 degrees angle of attack.

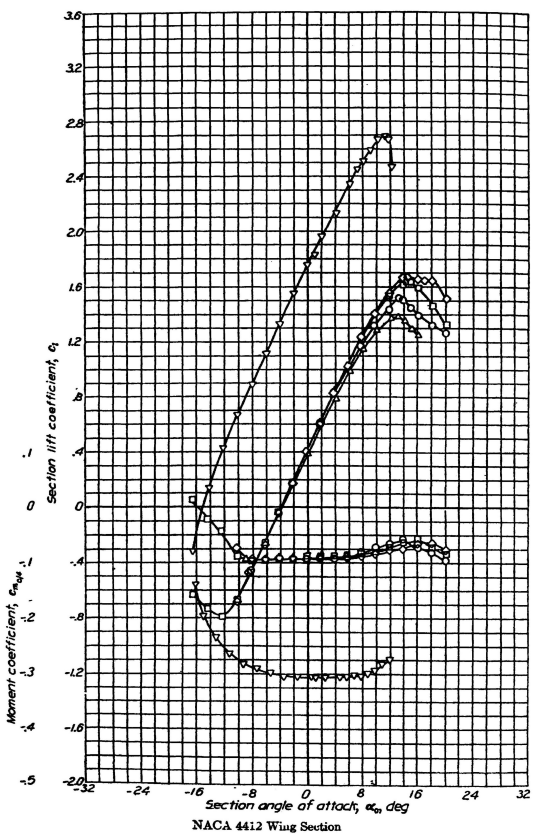
Figure 3: Smoke pathline flow visualizations of a NACA 0012 symmetric airfoil. The pathlines are not observed under the airfoil due to the shadow cast by the airfoil on the illuminating light.

References

- [1] I. Abbott and A. Von Doenhoff, *Theory of Wing Sections, Including a Summary of Airfoil Data*. Dover Books on Aeronautical Engineering Series, Dover Publications, 1959.

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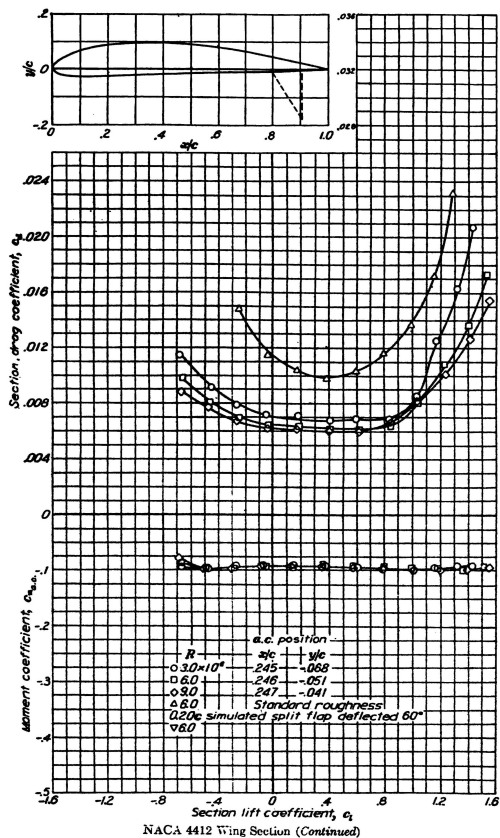
THEORY OF WING SECTIONS



(a) Airfoil without flaps.

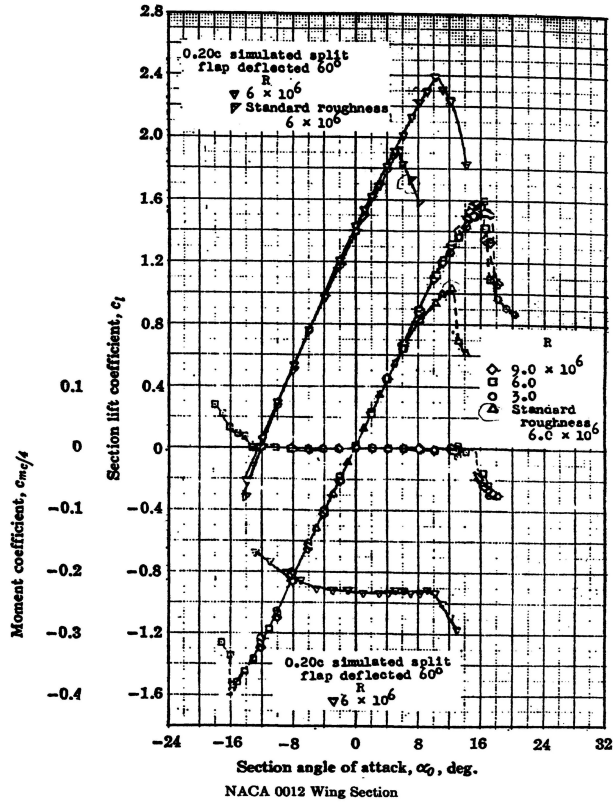
APPENDIX IV

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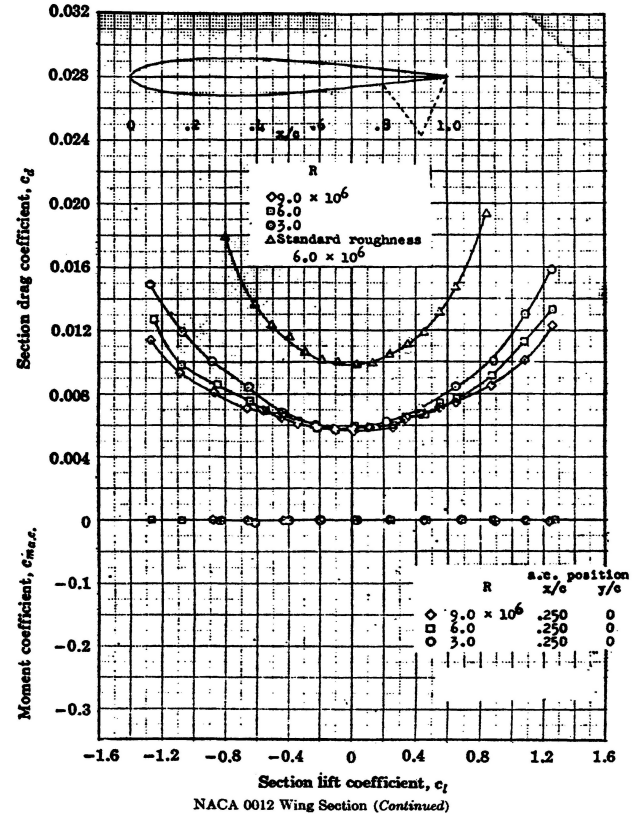


(b) Airfoil with flaps

Figure 4: Lift and drag coefficient data for high Reynolds number flow over an infinite wing section for NACA 4412. This section is not in the present test matrix and is included as a reference for the effect of camber.



(a) Airfoil without flaps.



(b) Airfoil with flaps

Figure 5: Lift and drag coefficient data for high Reynolds number flow over an infinite wing section for NACA 0012.

Appendix A Data Recording Sheets

Five airfoil sheets covering the 30 required trials, plus one sheet for the calibration mounting sting. The printed angles of attack assume the standard offset angle $\phi_0 = 10^\circ$; if a model's offset angle differs, cross out the printed values and enter the angles recomputed from (1).

Date: _____ Group: _____

NACA 0012 *Baseline, symmetric* Span: _____ Chord: _____ ϕ_0 : _____

α	θ	Wind off		Wind on		
		A (N)	N (N)	Manom.	A (N)	N (N)
-5°	-15°					
0°	-10°					
5°	-5°					
10°	0°					
15°	+5°					
20°	+10°					

NACA 2212 *Camber at 0.2c* Span: _____ Chord: _____ ϕ_0 : _____

α	θ	Wind off		Wind on		
		A (N)	N (N)	Manom.	A (N)	N (N)
-5°	-15°					
0°	-10°					
5°	-5°					
10°	0°					
15°	+5°					
20°	+10°					

NACA 2412 *Camber at 0.4c* Span: _____ Chord: _____ ϕ_0 : _____

α	θ	Wind off		Wind on		
		A (N)	N (N)	Manom.	A (N)	N (N)
-5°	-15°					
0°	-10°					
5°	-5°					
10°	0°					
15°	+5°					
20°	+10°					

Appendix A, continued

Date: _____ Group: _____

NACA 2424 Thickness 24%Span: _____ Chord: _____ ϕ_0 : _____

α	θ	Wind off		Wind on		
		A (N)	N (N)	Manom.	A (N)	N (N)
-5°	-15°					
0°	-10°					
5°	-5°					
10°	0°					
15°	+5°					
20°	+10°					

Fifth airfoil: _____ Span: _____ Chord: _____ ϕ_0 : _____

α	θ	Wind off		Wind on		
		A (N)	N (N)	Manom.	A (N)	N (N)
-5°	-15°					
0°	-10°					
5°	-5°					
10°	0°					
15°	+5°					
20°	+10°					

Calibration mounting sting No airfoil; tare run

Sting angles only

α	θ	Wind off		Wind on		
		A (N)	N (N)	Manom.	A (N)	N (N)
—	-15°					
—	-10°					
—	-5°					
—	0°					
—	+5°					
—	+10°					

Notes / anomalies observed:
